

EKG Design Project

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Introduction

The goal of this project is to build a compact ECG system that can capture the heart's electrical signals, clean them with a low-power analog front end, digitize them with a PIC24 microcontroller, and display the waveform on a laptop in a scrolling, roll-chart style. The system needs to clearly display the main features of an ECG signal, such as the P-wave, QRS complex, and T-wave, so the heartbeat can be easily recognized and analyzed.

Our general approach is to create a stable, low-noise signal chain that operates from small dual-battery supplies and stays within the PIC24's ADC limits. We're using modern precision parts like the INA333 instrumentation amplifier and the OPA333 op-amp to get the gain, low offset, and high common-mode rejection needed for microvolt-level ECG signals. Overall, the design focuses on keeping the circuit simple, reliable, and easy to interface with the digital processing and laptop display.

An ECG measures the electrical signals in the millivolt range produced as the heart goes through each beat. These signals form a repeating pattern made up of the P-wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T-wave (ventricular repolarization). Because these signals are only a few millivolts, precise amplification and filtering are required before they can be digitized. Older ECG circuits often used amplifiers such as the AD620. However, newer devices, such as the INA333 and OPA333, offer better performance for low-voltage, battery-powered designs. By combining these components with the PIC24's ADC and a laptop-based display, we can create a small, effective system for real-time ECG monitoring.

Prior Work

Circuit schematics for EKGs are super easy to find online with a quick Google search. The design we're using is based on a common schematic that uses an AD620A instrumentation amp along with AD705J op-amps. We're swapping those parts out for newer amplifiers, which means we have to redo a bunch of the resistor and capacitor values, but the overall layout stays pretty much the same.

A lot of ECG front-end circuits have already been done in textbooks, app notes, and old lab projects, so there's plenty of guidance on what a "standard" setup looks like. Older designs almost always used the AD620 because it was simple and had solid CMRR for its time. The problem is that newer research points out that the AD620 isn't great for low-voltage, battery-powered systems. It needs higher supply rails, burns more power, and has more offset and drift than modern parts.

Most recent ECG designs use low-power instrumentation amps like the INA333, INA326, or AD8237. These are built specifically for biopotential measurements, so they give you high CMRR, low noise, and rail-to-rail operation—basically everything you need when you're trying to amplify signals that are only a few millivolts. A lot of reference circuits also throw in a precision op-amp like the OPA333 for buffering, filtering, or shifting the signal to mid-supply before feeding it into an ADC.

Another trend is keeping the analog filtering super minimal. Most designs just use simple high-pass and low-pass filters to keep the signal in the ECG bandwidth and then clean everything else up digitally. It makes the analog section way easier to build and gives you more flexibility once the microcontroller takes over.

Overall, the existing designs point toward using a low-power instrumentation amp, a precision buffer stage, and a mid-rail bias to interface with a microcontroller's ADC. Our design is basically following that same approach, but we have to tweak it to match the PIC24's ADC range and the ± 3.3 V supply rails.

Contributions

Danny - LT Spice circuit design, built physical circuit, debugged circuit, designed altium schematic and layout, wrote Design Proposal

Danyal - Researched and found adequate chips, assisted LT Spice circuit design, built physical circuit, wrote Design Proposal

Henry - LT Spice circuit design, built a physical circuit, wrote Design Proposal

Samuel - ECE Depot coordinator, assisted LT Spice circuit design, built physical circuit, wrote Design Proposal

Project Design Specification

The objective of this project is to capture a patient's heartbeat using electrodes connected to the patient that generate voltage signals for the system. The system will then take the small input signal of the patient's heartbeat and feed it into an instrumentation amplifier with a gain of 7. This will allow any noise to be detected and filtered using a high-pass filter with a cutoff frequency of 0.3Hz. The amplified and filtered signal will then be amplified again using an operational amplifier with a gain of 143. This will result in a total gain of 1V/mV for the heartbeat signal, filtering any

unwanted noise. Additionally, the patient's body acts as an antenna for noise throughout the room, which the system treats as a common-mode signal. To counteract this signal, the use of another operational amplifier with an inverting unity gain relative to the common mode signal is fed back to the patient.

Specifications

- Ability to read signals in the mV range.
- Overall gain of ~ 1000 .
- Low power consumption at 3V.
- Dual power supply, +3V and -3V.
- Cancellation of common mode signal with right leg drive.
- Input to ADC must be between 0 and 3V.

The goal of this project is to build a full low-power ECG system that can pick up biopotential signals from standard electrodes, run them through a modern analog front end, digitize everything with the PIC24's ADC, and then show the waveform on a laptop in a scrolling, roll-chart style; basically like a simplified clinical ECG monitor.

The main engineering challenge is dealing with the fact that ECG signals are tiny, like microvolt-level tiny, and they get buried under noise, motion artifacts, and common-mode junk. On top of that, the whole thing has to run off $\pm 3\text{V}$ batteries and still output something that fits inside the PIC24's 0.2–3.1 V ADC range. Since the old AD620 isn't great for low-voltage setups, we're switching to newer low-power parts: the INA333 as the instrumentation amp and the OPA333 as the precision buffer. These give us the high CMRR, low noise, and rail-to-rail performance we need without the AD620's power and offset issues.

The final system needs to reliably capture clean ECG waveforms, clear P-waves, QRS complexes, T-waves, and hand them off to the digital side so the laptop can plot everything in real time.

Proposed Solution

Overview:

Ultimately, our team's proposed solution consists of connecting two electrodes to a person's body to record the electrical activity responsible for the heartbeat, and the $\sim 1\text{--}3\text{mV}$ pulsing heartbeat signal from the electrodes will be displayed on a PC. The

electrode connected to the right leg is connected to the OPA333, which acts as an inverting amplifier, canceling the common mode, of the output signal from the heart's beating electrical signal is passed through the INA333, an instrumentation amplifier. The reason we use the OPA333 as the right leg drive is for its high gain with a low supply voltage, making it a really good amplifier when amplifying the output of the INA333 to successfully supply the PIC24. The noise output by the instrumentation amplifier, INA333, undergoes a high-pass filter to get rid of low-frequency noise, adjusting to a reasonable RC value. The high-pass filter also gets rid of the small but non-zero offset voltage at the output of the op amp. Then, the signal is amplified by the OPA333 with a gain of $100\frac{V}{V}$. Before the signal from the high-pass filter reaches the PIC24 ADC, The signal must be offset so that it is in the range of the ADC, which is from 0V to $V_{DD}=3V$. A final OPA333 op amp is used as a summing amplifier between the signal and a 1.5V DC offset (created by a voltage divider), to get a final signal that swings through most of the ADC range, but leaving some headroom.

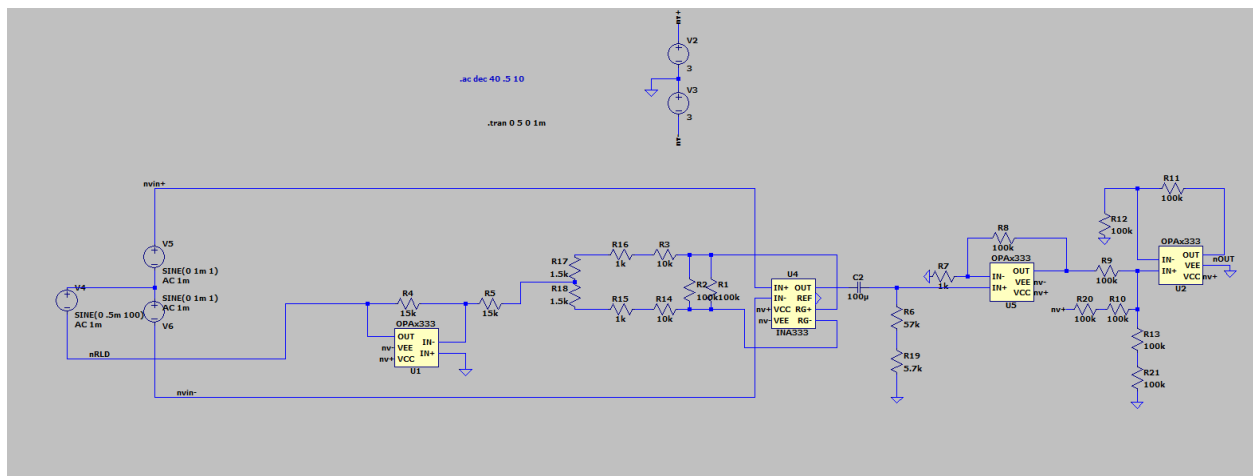


Figure 1: LTspice schematic of our proposed physical circuit.

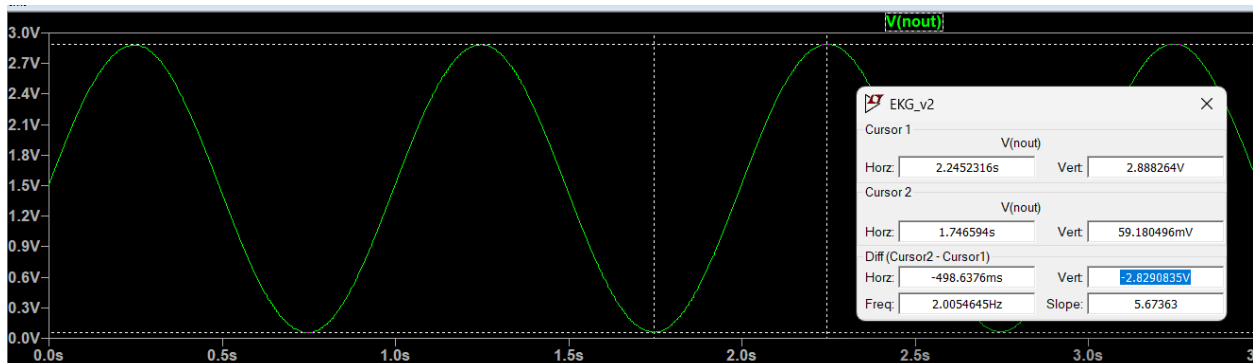


Figure 2: LTspice simulated output. Overall gain of 1414.5 (2 mVpp input) and dc offset of 1.5V to put the output in the 0-3V range to be read by the ADC.

Budget

The total cost for this project would come out to be \$40.31. This does not include our PCB or any holding unit; this also does not include the cost of production. All cost estimates are based on non-bulk, single-quantity pricing from Digi-Key.

For bulk orders, we found that the Digi-Key prices reduced by roughly 20%. So our estimated bulk cost would come out to be 32.00\$.

Bill

Component	Description	Qty	Unit Cost	Total Cost
INA333	Instrumentation amplifier	1	\$4.54	\$4.54
OPA333	Op amp	3	\$2.35	\$7.05
PIC24 Microcontroller	Microcontroller	1	\$8	\$8.00
Resistors		25	\$0.10	\$2.50
Capacitors		1	\$0.50	\$0.50
ECG connectors		3	\$5	\$15.00
Battery holders		2	\$1.36	\$2.72
Wires/Jumpers		25	\$0	\$0.00
Total				\$40.31

Conclusion

In conclusion, the main idea of this project is narrowed to being able to design a system that can act as an ECG by practicing schematic design, LT Spice Simulation, Hardware Troubleshooting, and ADC microcontroller programming. Furthermore, using the OPA333 as the right leg drive to act as an inverting amplifier, as well as the last stage to maintain a gain of 143 to supply the PIC24, and the INA333 behaves as a DC coupling amplifier. Being able to use the ADC in the PIC24 to digitize the analog circuit that we simulated and assembled is quite incredible for the technology that can read such a signal that is later converted into a readable pulsing heartbeat. By the end of the project, our goal is to assemble our simulated circuit, program the PIC24 to read the

ADC signals to output on an LCD, and make sure the noise of the signal will not have issues during the demonstration.

During the hardware setup we decided to use the INA333 and OPA33 components to reduce noise distortions and have a larger differential gain (higher CMRR). Altogether, the cost for making this design with considering extra chips ordered through Digi Key cost only \$39 and now is easy to integrate into future designs.

References

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